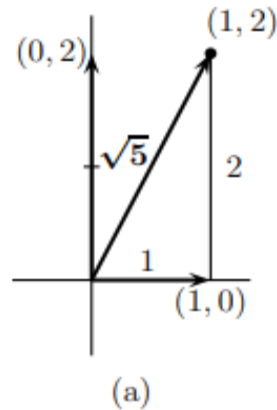


Orthogonal vectors and subspaces

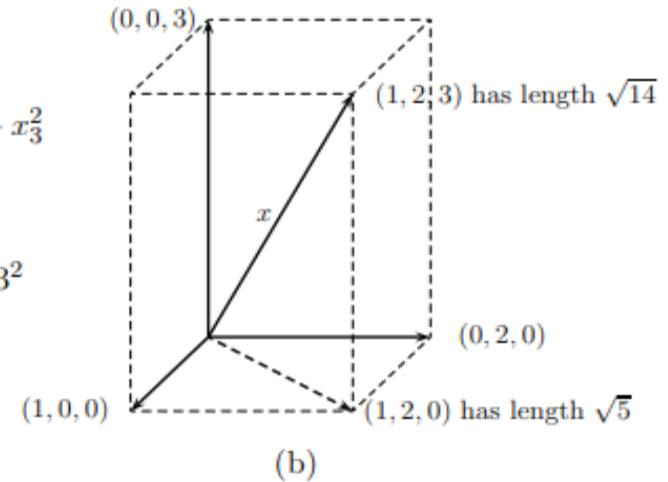
Length of a vector: It is denoted by $\|x\|$, the square of the length was given a long time ago by Pythagoras: $\|x\|^2 = x_1^2 + x_2^2$.



$$\|x\|^2 = x_1^2 + x_2^2 + x_3^2$$

$$5 = 1^2 + 2^2$$

$$14 = 1^2 + 2^2 + 3^2$$



So, in 3 dimensions $\|x\|^2 = x_1^2 + x_2^2 + x_3^2$ and length of $x = (x_1, x_2, x_3)$ is $\|x\| = \sqrt{x_1^2 + x_2^2 + x_3^2}$.

The extension to $x = (x_1, \dots, x_n)$ in n dimensions is immediate. By Pythagoras $n - 1$ times, the length $\|x\|$ in R^n is the positive square root of $x^T x$, i.e., $\|x\|^2 = x_1^2 + x_2^2 + \dots + x_n^2 = x^T x$.

Orthogonal vectors: Two vectors x and y are orthogonal if $x^T y = 0$ i.e., dot product of x and y is 0.

Proof: For right triangle,

$$\|x\|^2 + \|y\|^2 = \|x + y\|^2$$

$$x^T x + y^T y = (x + y)^T (x + y) = x^T x + x^T y + y^T x + y^T y$$

$$x^T y = 0$$

If $x = 0$, then x is orthogonal to every vectors.

Example : $(2, 2, -1)$ is orthogonal to $(-1, 2, 2)$. As $\begin{bmatrix} 2 & 2 & -1 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} = 0$

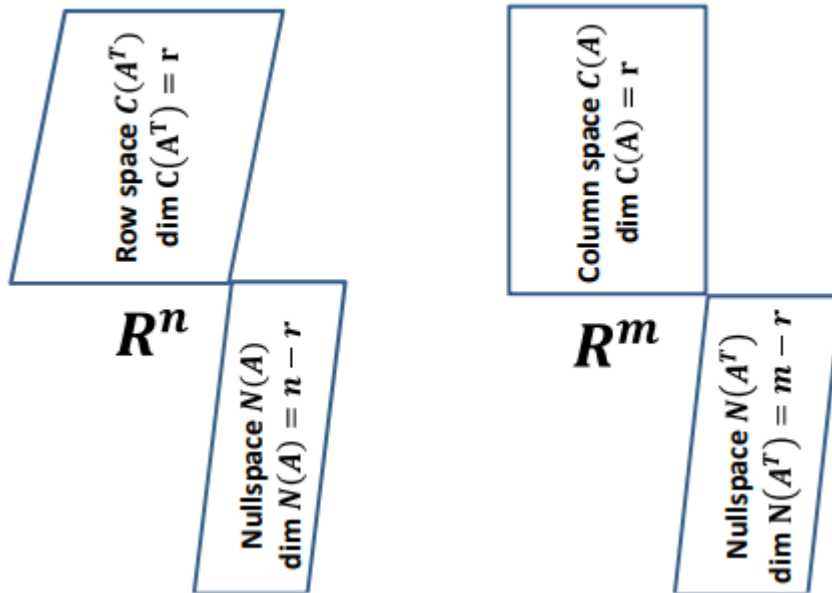
Orthogonal vectors and subspaces

Orthogonal Subspaces: Two subspaces are called orthogonal subspaces if every vector in one subspace must be orthogonal to every vector in the other subspace.

Subspaces of 2D-Plane: I. $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ II. Line passing through $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ III. Plane itself (i.e., R^2).

- Line through the $\begin{bmatrix} 0 \\ 0 \end{bmatrix} \perp R^2$. x
- Line through the $\begin{bmatrix} 0 \\ 0 \end{bmatrix} \perp \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. v
- Two lines through the $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$. $[(cx, 0), (0, dy)]$

Fundamental theorem of orthogonality: The row space is orthogonal to the nullspace (in R^n). The column space is orthogonal to the nullspace of A^T (in R^m).



$$Ax = \begin{bmatrix} \cdots & \text{row 1} & \cdots \\ \cdots & \text{row 2} & \cdots \\ \vdots & \vdots & \vdots \\ \cdots & \text{row } m & \cdots \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Orthogonal vectors and subspaces

Proof:

Suppose x is a vector in the nullspace. Then $Ax = 0$, can be written as:

$$Ax = \begin{bmatrix} \cdots & \text{row 1} & \cdots \\ \cdots & \text{row 2} & \cdots \\ \vdots & \vdots & \vdots \\ \cdots & \text{row } m & \cdots \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

This shows that x is orthogonal to every row of A , i.e., $(\text{row}_i)^T x = 0$.

Moreover, $c_i(\text{row}_i)^T x = 0$, and $c_j(\text{row}_j)^T x = 0$.

This implies, $c_i(\text{row}_i)^T x + c_j(\text{row}_j)^T x = 0$.

So, all the linear combinations of rows of A are orthogonal to x . Thus, row space $\perp$ nullspace.

Similarly, $A^T x = 0$, can be written as:

$$\begin{bmatrix} \cdots & \text{col 1} & \cdots \\ \cdots & \text{col 2} & \cdots \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdots & \text{col } n & \cdots \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_m \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \cdot \\ \cdot \\ 0 \end{bmatrix}.$$

This shows that x is orthogonal to every column of A , i.e., $(\text{col}_i)^T x = 0$.

Moreover, $c_i(\text{col}_i)^T x = 0$, and $c_j(\text{col}_j)^T x = 0$.

This implies, $c_i(\text{col}_i)^T x + c_j(\text{col}_j)^T x = 0$.

So, all the linear combinations of columns of A are orthogonal to x . Thus, column space $\perp$ nullspace of A^T .

Orthogonal vectors and subspaces

Example: Show that row space $\perp$ nullspace and column space $\perp$ null space of A^T for matrix $A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \\ 3 & 9 \end{bmatrix}$.

$A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \\ 3 & 9 \end{bmatrix} \longrightarrow U = \begin{bmatrix} 1 & 3 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$, This shows that A is a rank 1 matrix and its row space is a 1D line

multiple of $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, which is subspace of R^3 . Its column space is also a 1D line multiple of $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, which is subspace of R^3 .

$Ax = 0 \longrightarrow Ux = 0$, for $x = c \begin{bmatrix} -3 \\ 1 \end{bmatrix}$. So, nullspace of A is a line of multiple of $\begin{bmatrix} -3 \\ 1 \end{bmatrix}$ in R^2 . Since $\begin{bmatrix} 1 & 3 \end{bmatrix} \begin{bmatrix} -3 \\ 1 \end{bmatrix} = 0$, nullspace and row space are perpendicular in R^2 .

$A^T x = 0 \longrightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$. So nullspace of A^T is $c \cdot \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + d \cdot \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$, which is a plane in R^3 . Since $\begin{bmatrix} -5 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 0$, $\begin{bmatrix} -7 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 0$, nullspace and column space are perpendicular in R^3 .

➤ The nullspace is the orthogonal complement of the row space in R^n .

➤ The nullspace of A^T is the orthogonal complement of the column space in R^m .

Projection

Let A be a rectangular matrix. $A^T A$ is invertible exactly if A has independent columns.

Example: $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$, now $A^T A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 8 \\ 8 & 30 \end{bmatrix}$, which is invertible.

If $A = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix}$, then $A^T A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 12 \end{bmatrix}$, which is singular matrix.

Projection: Want to find a point p on the subspace (line $\vec{a}$) that closest to b .

The projection p is some multiple of $\vec{a}$, i. e. $p = ax$.

Since, a is perpendicular to $b - p$,

$$a^T(b - ax) = 0$$

$$a^T ax = a^T b$$

$$x = \frac{a^T b}{a^T a}$$

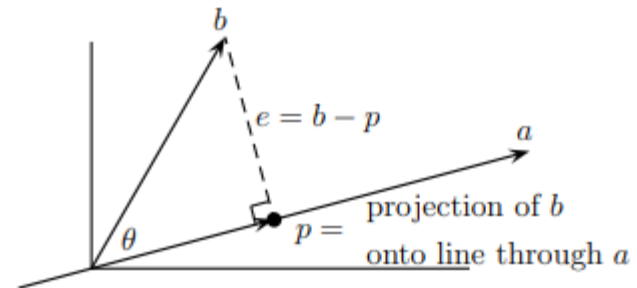
So, $p = ax = a \frac{a^T b}{a^T a} = Pb$, where $P = \frac{aa^T}{a^T a}$ is projection matrix.

If $b = 2b$, then $p = 2p$. If $a = 2a$, then $p = p$.

The projection matrix P is a $m \times m$ of rank 1 matrix. Thus $C(P) = \text{line through } a$.

Properties of P :

- I. Symmetric, i.e., $P^T = P$
- II. Second time projection of p is p only, i. e., $P^2 = P$

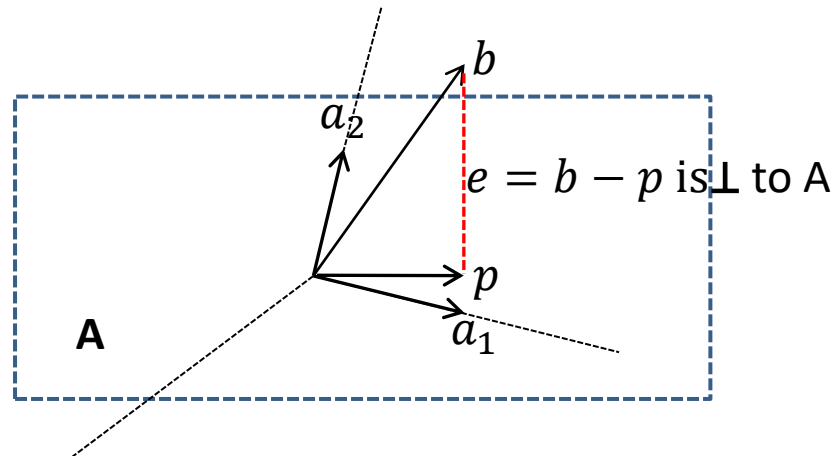


The projection p is the point (on the line through a) closest to b .

Projection

Why Projection?

Let A is an $m \times n$ matrix and $m > n$. In this case, for $Ax = b$, the number of equations is larger than the number of unknown, i.e., the rank of A is $\leq n$. So, for a large number of b have no solution. We have to find out the best solution of $Ax = b$.



Because $Ax = b$ may have no solution when $m > n$. I solve the closest problem $A\hat{x} = p$, where p is the projection of b onto column space.

Plane A formed by independent columns a_1 , and $a_2 = C(A) = \begin{bmatrix} \vdots & \vdots \\ a_1 & a_2 \\ \vdots & \vdots \end{bmatrix}$,

So the projection $p = \hat{x}_1 a_1 + \hat{x}_2 a_2 = A\hat{x}$.

Now the problem is to find out $\hat{x}$ for $p = A\hat{x}$, where $b - A\hat{x}$ is $\perp$ to the plane A , i.e., $b - A\hat{x}$ is $\perp$ to a_1 , and a_2 in A .

Projection

So, $a_1^T(b - A\hat{x}) = 0$ and $a_2^T(b - A\hat{x}) = 0$. This means $\begin{bmatrix} a_1^T \\ a_2^T \end{bmatrix} [b - A\hat{x}] = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$,

So,

$$A^T(b - A\hat{x}) = 0$$

$$A^T A \hat{x} = A^T b$$

$$\hat{x} = (A^T A)^{-1} A^T b$$

Now, projection $p = A\hat{x} = A(A^T A)^{-1} A^T b$. So, the projection matrix $P = A(A^T A)^{-1} A^T$.

If A is an invertible square matrix, then $P = A(A^T A)^{-1} A^T = A A^{-1} ((A^T)^{-1} (A^T)) = I$, So $p = b$.

Properties of P :

I. Symmetric, i.e., $P^T = P$

$$P^T = (A(A^T A)^{-1} A^T)^T = (A^T)^T ((A^T A)^{-1})^T A^T = A(A^T A)^{-1} A^T = P$$

I. Second time projection of p is p only, i.e., $P^2 = P$

$$P^2 = A(A^T A)^{-1} A^T A(A^T A)^{-1} A^T = A(A^T A)^{-1} A^T = P$$

Case I: b is $\perp$ to $C(A)$

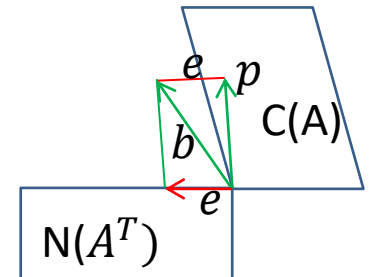
So, b is in nullspace of A^T , i.e., $A^T b = 0$

$$p = Pb = A(A^T A)^{-1} A^T \cdot b = A(A^T A)^{-1} 0 = [0]$$

Case II: b is in $C(A)$

So, b is in the column space of A , i.e., $b = Ax$

$$p = Pb = A(A^T A)^{-1} A^T \cdot b = A(A^T A)^{-1} (A^T A)x = Ax = b$$

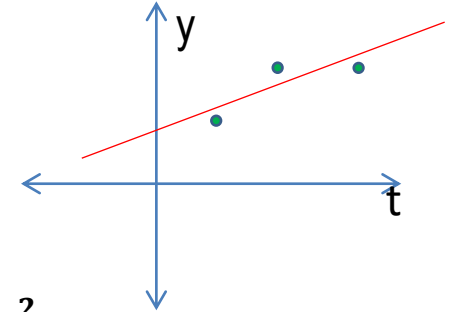


Least squares

Let $(1, 1)$, $(2, 2)$ and $(3, 2)$ be the three points in 2-D space and we want to fit a line $y = mt + c$ through these three points. The equation of line passing through $(1, 1)$, $(2, 2)$ and $(3, 2)$ is $m + c = 1$, $2m + c = 2$, and $3m + c = 2$, respectively. So, the linear system $Ax = b$ corresponding to this problem will be:

$$Ax = b$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} c \\ m \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$



Since, b is not in $C(A)$, The best solution will be

$$\text{Minimize } \|Ax - b\|^2, \text{ i.e., minimize } \|e\|^2 = e_1^2 + e_2^2 + e_3^2$$

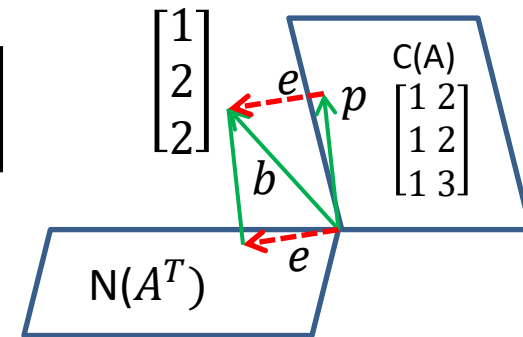
So, we need to find $\hat{x} = \begin{bmatrix} c \\ m \end{bmatrix}$, such that the projection $p = A\hat{x}$ of b onto the column space $C(A)$.

$$\text{We know that } A^T(b - A\hat{x}) = 0 \Rightarrow A^T A\hat{x} = A^T b$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} c \\ m \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \begin{bmatrix} c \\ m \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} 3c + 6m &= 5 \\ 6c + 14m &= 11 \end{aligned}$$



The above system can be solve using Gaussian elimination method as follow:

$$\begin{bmatrix} 3 & 6 & 5 \\ 6 & 14 & 11 \end{bmatrix} = \begin{bmatrix} 3 & 6 & 5 \\ 0 & 2 & 1 \end{bmatrix}, \text{ So } 2m = 1 \rightarrow m = 1/2 \text{ and } 3c + 6m = 5 \rightarrow c = 2/3$$

Least squares

So, $\hat{x} = \begin{bmatrix} c \\ m \end{bmatrix} = \begin{bmatrix} 2/3 \\ 1/2 \end{bmatrix}$. Hence, the best line closest to all the three points is $y = 1/2 t + 2/3$.

The projection $p = A\hat{x} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 2/3 \\ 1/2 \end{bmatrix} = \begin{bmatrix} 7/6 \\ 5/3 \\ 13/6 \end{bmatrix}$ | *for (1, y), y = 7/6, for (2, y), y = 5/3, for (3, y), y = 13/6*

The error $e = (b - A\hat{x}) = \begin{bmatrix} -1/6 \\ 1/3 \\ -1/6 \end{bmatrix}$ and $b = p + e = \begin{bmatrix} 7/6 \\ 5/3 \\ 13/6 \end{bmatrix} + \begin{bmatrix} -1/6 \\ 1/3 \\ -1/6 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. p and e are perpendicular.

Theorem: If A has independent columns then $A^T A$ is invertible.

Proof: Let $A^T A x = 0$

So, $x^T A^T A x = 0$ (multiply x^T both side)

$(Ax)^T (Ax) = 0$, This shows that length of vector Ax is 0, which implies that $x = 0$ if A has independent columns.

So, $A^T A x = 0$, when $x = 0$, i.e., $A^T A$ is invertible if A has independent columns.

Orthogonal Bases

In an orthogonal basis, every vector is perpendicular to every other vector. The vectors $q_1, q_2, \dots, q_n$ are orthonormal basis if

$$q_i^T q_j = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

A matrix with orthonormal columns is called orthonormal matrix Q .

$$\text{Let } Q = \begin{bmatrix} \vdots & \vdots & \dots & \vdots \\ q_1 & q_2 & \dots & q_n \\ \vdots & \vdots & \dots & \vdots \end{bmatrix}, \quad Q^T Q = \begin{bmatrix} \dots & q_1 & \dots \\ \dots & q_2 & \dots \\ \dots & \dots & \dots \\ \dots & q_n & \dots \end{bmatrix} \begin{bmatrix} \vdots & \vdots & \dots & \vdots \\ q_1 & q_2 & \dots & q_n \\ \vdots & \vdots & \dots & \vdots \end{bmatrix} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 1 \end{bmatrix} = I$$

If Q is square matrix, then $Q^T Q = I$, tell us $Q^T = Q^{-1}$.

Example of orthonormal matrices:

I. Permutation matrices $Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix},$

II. $Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

III. $Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix},$

IV. $Q = \frac{1}{3} \begin{bmatrix} 1 & -2 \\ 2 & -1 \\ 2 & 2 \end{bmatrix}$

Projections onto the column space formed by Q

We know that $P = A(A^T A)^{-1} A^T$

$$\begin{aligned} &= Q(Q^T Q)^{-1} Q^T \\ &= Q I Q^T \\ &= Q Q^T \end{aligned}$$

If Q is square matrix then column space of Q is the whole space, so the projection remain same. In this case

$$P = Q Q^T = Q Q^{-1} = I.$$

Properties of projection matrix P :

$P = Q Q^T$ is symmetric.

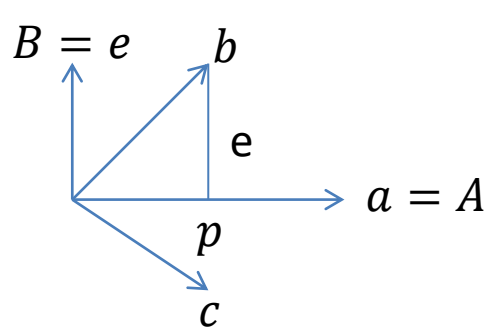
$$P^2 = Q Q^T \cdot Q Q^T = Q(Q^T Q) Q^T = Q Q^T = P$$

Form projection we know that $A^T A \hat{x} = A^T b$. If A is replace by Q then $Q^T Q \hat{x} = Q^T b$

This implies $\hat{x} = Q^T b$, i.e., $\hat{x}_i = q_i^T b$.

Gram-Schmith method

Gram-Schmith Process for making orthonormal matrix:



a, b two independent vectors

Get orthogonal vectors A, B

Orthonormal vectors
 $q_1 = \frac{A}{\|A\|}, q_2 = \frac{B}{\|B\|}$

Let $A = a$, and p is the projection of b onto A . According to the projection, $B = e = b - Ax = b - A(A^T A)^{-1} A^T b$.

Check A and B are orthogonal: $A^T(b - A(A^T A)^{-1} A^T b) = A^T b - A^T A(A^T A)^{-1} A^T b$
 $= A^T b - A^T b$
 $= 0$

a, b, c three independent vectors

Get orthogonal vectors A, B, C

Orthonormal vectors
 $q_1 = \frac{A}{\|A\|}, q_2 = \frac{B}{\|B\|}, q_3 = \frac{C}{\|C\|}$

$A = a,$
 $B = b - A(A^T A)^{-1} A^T b,$
 $C = c - A(A^T A)^{-1} A^T c - B(B^T B)^{-1} B^T c,$

Gram-Schmith method

Example: Find the orthonormal matrix of $\begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 2 \end{bmatrix}$.

$$\text{Let } A = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \text{ now } B = b - Ax = b - A(A^T A)^{-1} A^T b = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - A \left([1 \ 1 \ 1] \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right)^{-1} \left([1 \ 1 \ 1] \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - \frac{3}{3} A = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}.$$

$$\text{So, } q_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \text{ and } q_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}. \text{ Hence the orthonormal matrix will be } \begin{bmatrix} 1/\sqrt{3} & 0 \\ 1/\sqrt{3} & -1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{2} \end{bmatrix}.$$

Example: Find the orthonormal matrix of $\begin{bmatrix} 1 & 1 & 2 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$.

$$\text{Let } A = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \text{ now } B = b - Ax = b - A(A^T A)^{-1} A^T b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - A \left([1 \ 0 \ 1] \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right)^{-1} \left([1 \ 0 \ 1] \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \frac{1}{2} A = \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}.$$

$$C = c - Ax - Bx = c - A(A^T A)^{-1} A^T c - B(B^T B)^{-1} B^T c = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - A(2)^{-1}(2) - B \left(\frac{1}{2} \right)^{-1} (1) = c - A - 2B = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

$$\text{So, } q_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \text{ and } q_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \text{ and } q_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}. \text{ Hence the orthonormal matrix will be } \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \end{bmatrix}.$$